Individual Coursework Submission Form

Specialist Masters Programme

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Introduction

To support a fair and data-driven restructuring strategy, the analysis began with the previous closure dataset to identify patterns in voluntary attrition. Exploratory insights revealed that employees with lower tenure, income, and job satisfaction were more likely to accept a Rupture Conventionnelle Collective (RCC). After addressing outliers and cleaning the data, various machine learning models were evaluated. Logistic regression emerged as the most effective—offering high performance and clear interpretability—especially after applying SMOTE to correct class imbalance and tuning hyperparameters through GridSearchCV.

With the model finalized, the same methodology was applied to the Lyon office dataset. After consistent preprocessing, the model predicted each employee's likelihood of accepting an RCC. These probabilities were compiled into a CSV file, enabling precise, group-level planning for the next phase of optimization. This approach ensured that workforce decisions could be both evidence-based and aligned with organizational and legal considerations.

Task 3: Group Definition Strategy for RCC Offers

To ensure that RCCs are offered only to "objectively defined" employee groups, a descriptive grouping framework was designed, grounded in the key predictors of RCC acceptance. These groups are built using features that are both highly predictive of voluntary attrition and neutral in legal and ethical terms—a critical requirement under French labour law, which prohibits targeting individuals or subjective categories.

Feature Selection Strategy

The grouping strategy is informed by insights from Task 1, where logistic regression revealed the top factors influencing an employee's decision to accept an RCC. These features were selected not only for their statistical importance (based on model coefficients) but also for their business interpretability and domain relevance [Table1].

FEATURE	REASON FOR INCLUSION	DOMAIN INSIGHT
OVERTIME	Strongest predictor in the model	Reflects workload and burnout risk
NUMCOMPANIESWORKED	Indicator of job mobility	Highly mobile employees are more open to change
YEARSINCURRENTROLE	Tenure and role stability	Short tenure suggests weaker attachment
ENVIRONMENTSATISFACTION	Proxy for morale	Dissatisfied employees are more likely to leave
DEPARTMENT	Role type and strategic importance	Reflects functional exposure and criticality
JOBROLE_HR	Refines department classification	HR staff may be more sensitive to organizational change

Table1: Key Features Used for Strategic Group Definition and Their Relevance

These dimensions were combined to form strategic group labels such as *"High-Mobility Overtime | R&D New"* or *"Low-Mobility Standard | Sales Established"*. The logic ensures that group formation is grounded in observable characteristics rather than arbitrary criteria, making them objectively definable, explainable, and reproducible.

Balancing Complexity and Practicality

Initially, the combination of features produced over 15 unique groups. To maintain feasibility and prevent overly granular segmentation:

- Groups with fewer than five employees were merged into similar, larger ones.
- Tenure categories were collapsed by department to form broader labels like *"R&D New"* or *"HR Established"*.
- Satisfaction was only used to further divide overtime-heavy groups (those most at risk of leaving), preserving both accuracy and simplicity.

This yielded a manageable number of strategic groups that retain predictive power without being administratively burdensome.

Group Size Pros Cons

This table evaluates the trade-offs between small and large employee groups in the context of offering RCCs, balancing predictive power, fairness, and practical implementation.

Group	Pros	Cons
Smaller Groups	More precise targeting; better reflects nuanced risk profiles	Can appear overly tailored, raising legal concerns; harder to administer; risk of overfitting
Larger Groups	Easier to manage and defend legally; aligns with collective agreement spirit	Less precise; may blend high- and low-risk employees, reducing optimization effectiveness

Table 2: Pros and Cons of Group Size in RCC Planning

In workforce restructuring and HR planning, smaller groups often enable more accurate targeting by isolating distinct behavioral or risk traits—such as high burnout or low satisfaction. However, in regulated environments like France, overly specific groupings can raise concerns about discrimination or targeted redundancy, making them harder to defend legally and ethically.

Conversely, larger groups are easier to implement and are generally perceived as fairer under collective bargaining frameworks. Yet, they risk masking important variation—which can lead to missed opportunities or imbalanced workforce outcomes.

In this assignment, I addressed this trade-off by constructing medium-sized, logically aggregated groups using high-impact, business-relevant features. By merging very small segments and simplifying tenure categories, I ensured the grouping strategy was actionable, compliant, and predictive. This approach reflects real-world HR practices where data-driven planning must respect legal frameworks while delivering on strategic goals like cost reduction and workforce continuity.

Ensuring Non-Discriminatory Practices

To uphold fairness and legal defensibility in the grouping strategy, multiple safeguards were implemented to ensure that the approach aligns with ethical workforce planning and complies with French labour regulations, particularly the requirement that RCCs must be offered to objectively defined groups, not individuals.

1. **Departmental Balance within Groups**

From a business and regulatory perspective, one key fairness metric is to avoid forming groups that over-concentrate employees from any one department, which could unintentionally lead to indirect discrimination. In this assignment, it was ensured that no strategic group was composed of more than 80% of any single

department—be it Sales, R&D, or HR. This threshold helps protect departmental integrity and preserves critical operational capacity across functions, which is especially important in restructuring scenarios.
Moreover, in the Solver model, after RCC acceptance it was verified that, each department would retain at least 80% of its original headcount, aligning with the assignment's explicit constraints and broader workforce sustainability principles.

2. **Use of Legally Neutral, Business-Relevant Features**

Only role-specific and operationally relevant features were used to define groups. For example:

- OverTime status (workload)
- Job role and department (functional classification)
- Tenure in role (organizational attachment)
- Satisfaction levels (indicators of morale)

Crucially, no sensitive or protected attributes such as gender, age, marital status, nationality, or ethnicity were used, even though they may correlate with attrition risk. This decision reflects best practices in responsible AI and HR analytics, avoiding any form of bias or discriminatory profiling.

3. **Uniform Application of Criteria**

All employees were grouped using the same logic and feature set, regardless of department, seniority, or background. This consistency is critical in preventing selective targeting, a common risk in manual restructuring decisions.
By standardizing the grouping logic and making it data-driven and reproducible, the approach ensures transparency and equity, providing strong justification if challenged by employee representatives or labour authorities.

Task 4: Optimization Strategy for RCC Allocation

To support BAP in achieving an efficient and fair restructuring at SFB's Lyon office, an optimization model was formulated that determines which employee groups should be offered RCCs in order to minimize total severance cost, while respecting business constraints. This model was implemented in Excel using Solver.

Objective Function: Minimize Severance Cost

From an organizational standpoint, severance payouts represent a significant restructuring expense. The goal of this optimization is to minimize the expected total cost of RCCs offered across different employee groups. Each group has a known average monthly income, and RCC costs are assumed to be 3 months' salary for each expected leaver.

Expected leavers per group are calculated using:

$$\textit{Expected Leavers} = \textit{Group Size} \times \textit{RCC Acceptance Probability} \times \textit{RCC Offer Decision (binary)}$$

Severance cost per group is:

$$\textit{Expected Leavers} \times \textit{Average Monthly Income} \times 3$$

Minimizing the sum of these values ensures financial prudence while achieving operational goals.

Constraints: Legal and Operational Requirements

The model incorporates several critical constraints:

CONSTRAINT	RATIONALE
At least 40 employees must leave	Operational requirement to reduce workforce size
Minimum €3 million in annual salary savings	Ensures restructuring has substantial financial impact
Each department must retain ≤ 80% of its staff	Ensures balanced restructuring and prevents over-concentration of layoffs

Table 3: Optimization Constraints and Their Strategic Rationale

From a domain perspective, such constraints reflect typical safeguards embedded in collective bargaining agreements and restructuring policies. They help ensure that no department is unfairly depleted, and that the scale of reduction aligns with strategic goals.

These constraints were modeled in Solver using binary decision variables (1 if the group receives RCC offers, 0 otherwise), linked to group-level expected attrition and salary data.

Decision Variables: RCC Offer Allocation

The decision variable for each strategic group is binary:

- **1** if the group receives RCC offers
- **0** otherwise

These groups were defined earlier using objective, business-relevant features such as overtime, tenure, and department. By offering RCCs at the group level, the model complies with legal requirements that prohibit individual-level targeting.

Selected Groups

Group ID	***Group Description***	***Expected Leavers***	***Annual Savings***	***RCC Cost***
G6	High-Mobility Standard | R&D New	7.96	€512,391	€128,098
G5	High-Mobility Standard | R&D Established	5.05	€489,333	€122,333
G20	Low-Mobility Standard | Sales New	4.42	€311,560	€77,890
G13	Low-Mobility Overtime | Sales Established | High Sat.	3.20	€271,457	€67,864
G8	High-Mobility Standard | Sales New	3.30	€263,984	€65,996

These groups were selected based on their high attrition probabilities and favourable cost-to-savings ratios, ensuring efficient use of RCCs.

Task 5: Evaluation of the Prediction-and-Optimization Approach

The prediction-and-optimization framework used in this analysis combines machine learning with mathematical optimization to support workforce restructuring at SFB in a data-driven, scalable, and legally compliant manner. While this approach offers clear advantages in terms of precision and transparency, it is important to critically assess its limitations and the assumptions it relies upon.

Pros of the Prediction-and-Optimization Strategy

1. **Data-Driven and Objective Decision-Making**

   By basing RCC decisions on predicted probabilities of acceptance, the model avoids intuition-based or

subjective choices. This helps reduce personal bias, improves fairness, and supports defensibility under employment law.

2. **Efficient Resource Allocation**
   Optimization ensures that RCC offers are targeted to groups that will generate the most cost-effective outcomes (e.g., high attrition probability + low severance cost), helping BAP achieve savings goals without overspending.
3. **Scalability and Repeatability**
   Once trained and validated, the prediction model and optimization logic can be reapplied across different business units or timeframes, making it a valuable tool for strategic workforce planning.
4. **Legal Alignment**
   The group-based structure and exclusion of sensitive features ensure compliance with regulations that prohibit individual targeting and protect employees from discrimination.
5. **Uniform RCC Cost Assumption**
   The model assumes that all employees offered an RCC will receive the same severance compensation—three months of salary—regardless of role or seniority. This simplifies the cost calculations and reflects standard HR practice under collective agreements.

Cons and Limitations

1. **Reliance on Historical Patterns**
   The predictive model assumes that employees in the Lyon office will behave similarly to those in the previous closure. If organizational culture, communication, or economic factors differ, the predictions may not generalize well.
2. **Simplified Acceptance Behavior**
   The model treats acceptance probabilities as static and rational—e.g., an employee with a 0.6 likelihood will behave as part of an expected average. In reality, individual choices may deviate based on last-minute factors like peer influence or personal circumstances.
3. **No Feedback Loops or Dynamic Behavior**
   Once RCC offers are made, employees may influence each other's decisions (e.g., cascading exits within teams). This type of dynamic, emergent behavior is not accounted for in the model.
4. **Lack of Capacity Constraints or Interdependencies**
   The optimization does not model inter-group dependencies (e.g., removing a manager and their direct report together). This may lead to unintended organizational side effects if key roles are vacated simultaneously.

Key Assumptions and Their Plausibility

Assumption	***Realistic?***	***Comments***
Historical behavior predicts future behavior	Moderately	Depends on how similar Lyon is to the previous closure setting
Attrition probability estimates are accurate and stable	Reasonably	Model was validated and tuned, but may still have variance in real-world settings
Employees behave independently when making RCC decisions	Low	Peer and team dynamics can play a significant role
RCC cost is uniform across employees (3 months' salary)	High	Reasonable assumption in this context; simplifies optimization logic
All RCCs must be offered at group level, not individually	High	This is both a legal and modeling constraint in this case

Conclusion

While the prediction-and-optimization approach enables BAP to make structured, cost-efficient, and legally sound decisions, it is important to interpret the results as strategic guidelines rather than absolute prescriptions. To maximize effectiveness, these models should be used in conjunction with qualitative inputs from HR and leadership, and continuously refined based on feedback from actual implementation.